PHARMACOLOGY LAB 4(2026-27)

INTRAVENOUS FLUID THERAPY

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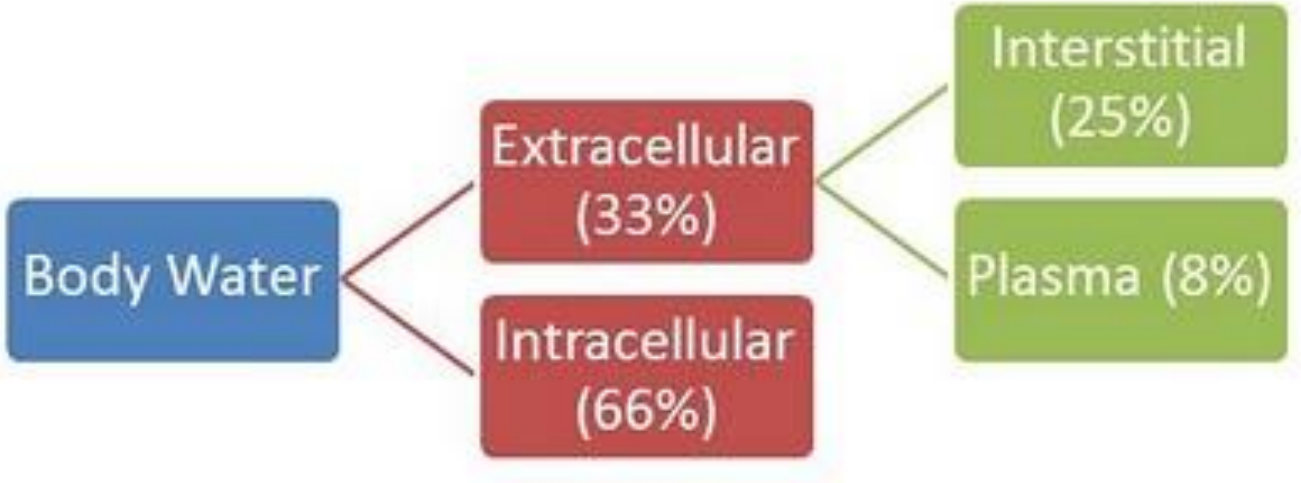
TOTAL BODY WATER

- Water is essential for life.
- Total body water (TBW), accounts for approximately 60% of total body weight.
- TBW is divided into intracellular and extracellular fluid.

ECF (EXTRACELLULAR FLUID)

- Accounts for 1/3 of the TBW

1. Interstitial compartment
2. Intravascular (plasma)
3. 3rd space



INDICATIONS FOR IV FLUID THERAPY

1-Fluid volume maintenance

2-Fluid volume replacement

3-Medication administration

4-TPN (total parenteral nutrition)

5- Emergency line


HDB3WG0E
Baxter
Zentralvenös
Mit Elektrolyten
Olimel
4,4 % E
Emulsion zur Infusion
HDB3WG0E
verwendbar bis
01-2023
Beutel
44,3
1140
960
1000 ml
21B15N51

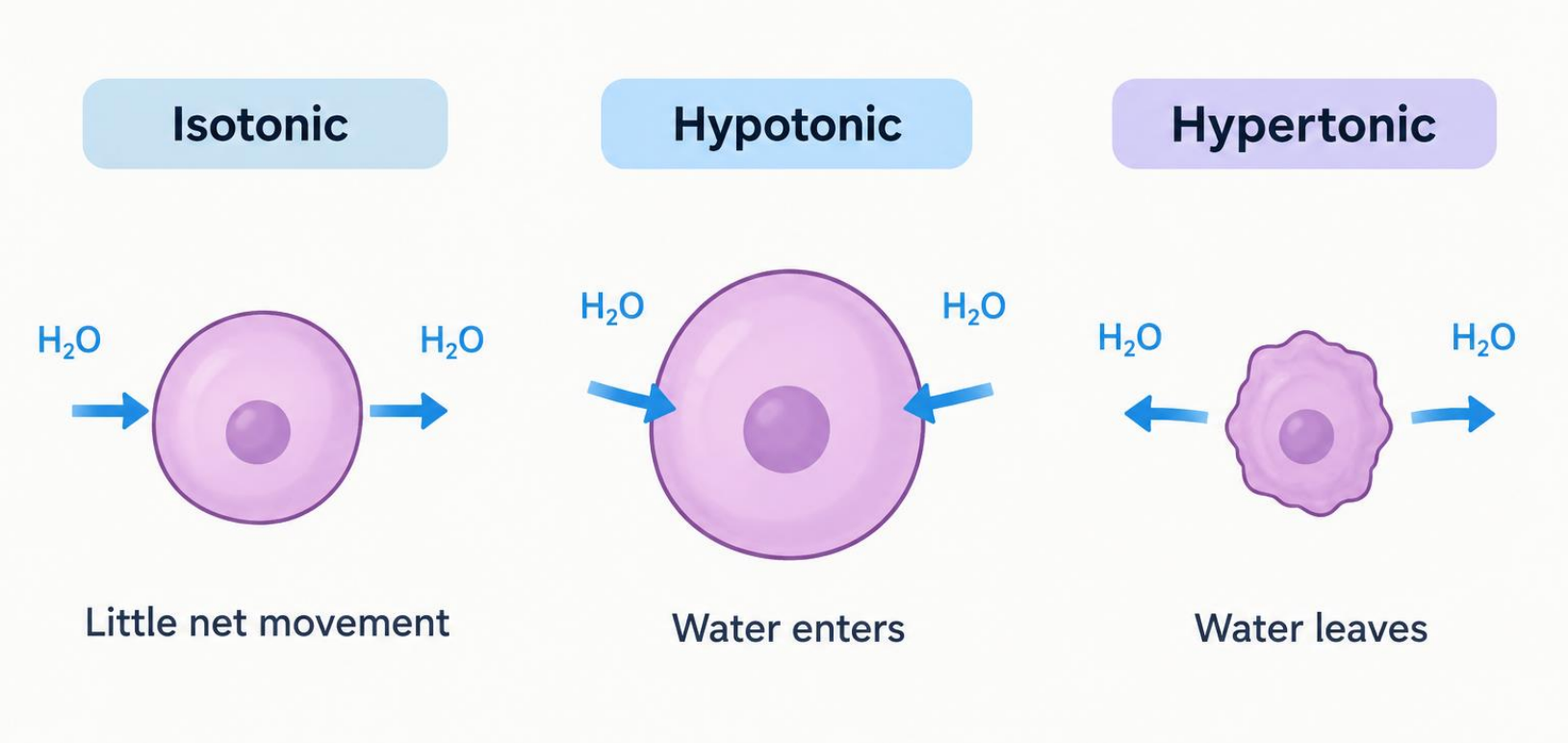
TYPES OF FLUIDS

The fluids used in clinical practice are classified into

1. Crystalloids (contain small molecules that flow easily across cell membranes)
2. Colloids (contain large molecules that don't pass cell membranes)
3. Blood products

TYPES OF CRYSTALLOID FLUIDS

I. Isotonic
II. Hypotonic
III. Hypertonic



I-ISOTONIC FLUIDS

Types of Isotonic fluids

A. 0.9% sodium chloride (0.9% NaCl).
B. Lactate Ringer's solution.
C. 5% Dextrose in water.
D. Ringer's solution.

- A solution is considered isotonic when the concentration of the particles (solutes) is similar to that of plasma.
- It doesn't move into cells but remains within the extracellular compartment thus increasing intravascular volume.

ISOTONIC FLUIDS: A. 0.9% SODIUM CHLORIDE (NORMAL SALINE)

- Simply salt water that contains only water , sodium and chloride
- It's called normal saline because the percentage of sodium chloride in the solution is similar to the concentration of sodium and chloride in the intravascular space.



ISOTONIC FLUIDS: A. 0.9% SODIUM CHLORIDE (NORMAL SALINE)

Indication of N.S:

1. To treat fluid volume deficit due to

- Bleeding
- Severe vomiting or diarrhea

2. Shock

3. Mild hyponatremia

ISOTONIC FLUIDS: B. RINGER'S LACTATE OR HARTMAN SOLUTION

- The electrolyte content is most closely related to the composition of the body's blood serum and plasma
- Contains sodium, potassium, calcium, chloride and lactate



ISOTONIC FLUIDS: B. RINGER'S LACTATE OR HARTMAN SOLUTION

Indications of R.L:

1. To replace G.I.T. fluid loss (vomiting or diarrhea).
2. Fluid loss due to burns and trauma.
3. Acute blood loss.

ISOTONIC FLUIDS: C. DEXTROSE 5%

- It is considered an isotonic solution, but when the dextrose is metabolized, the solution becomes hypotonic and causes fluid to shift into cells.



ISOTONIC FLUIDS: C. DEXTROSE 5%

Take care!

- D5W is not good for patients with **renal failure** or **cardiac problems** as it causes fluid overload
- Patients at risk for intracranial pressure shouldn't receive D5W since it could increase **cerebral edema.**

ISOTONIC FLUIDS: D. RINGER'S SOLUTION

- Like Ringer's lactate contains Na, K, Ca, Cl in similar amounts but not the lactate.

II- HYPOTONIC FLUIDS

- Compared with intracellular fluids , hypotonic solutions have a lower concentration of solutes (electrolytes).
- They cause fluids to shift from the intravascular space to both the intracellular and interstitial spaces.
- These will hydrate cells, although their use may deplete fluid within the circulatory system.

II- HYPOTONIC FLUIDS

Types of hypotonic solutions:

A. 0.45% sodium chloride
B. 0.33% sodium chloride
C. 0.2% sodium chloride
D. 2.5% dextrose in water

II- HYPOTONIC FLUIDS

- Hypernatremia
- Diabetic ketoacidosis
- Hyperosmolar hyperglycemic state (high blood sugar, dehydration)



II- HYPERTONIC SOLUTIONS

- Solution that have a high solute concentration.
- The osmotic pressure gradient draws water out of the intracellular space, increasing extracellular fluid volume, so they are used as **volume expanders**.

II- HYPERTONIC SOLUTIONS

Types of hypertonic solutions:

1. 3% sodium chloride (3% NaCl)

- In case of severe hyponatremia
- Patients with cerebral edema

2. 5% Dextrose with normal saline (D5NS)

COLLOID SOLUTIONS

These are used for **volume expansion**.

Examples:

1- 5% albumin (Human albumin solution) (natural).

2- Hydroxyethyl starches (synthetic).

I.V. FLUID ADMIXTURE

- An I.V. fluid admixture is the preparation of a pharmaceutical mixture of two or more drugs into a bottle of I.V. fluid.
- This is always done under the direction of a doctor, controlled and performed by a trained pharmacist.
- Drugs should not be added to blood, amino acid solutions and fat emulsions.

INCOMPATIBILITY

Incompatibility is a reaction between drugs that after mixing are no longer safe or effective for the patient.

Characteristics:

a) Color change

b) Hazy appearance

c) Precipitation

There are different types of incompatibles encountered in preparing IV admixture:

Types

1. **Physical incompatibilities**
2. **Chemical incompatibilities**
3. **Therapeutic incompatibilities**
4. **Drug IV container incompatibilities**

1-PHYSICAL INCOMPATIBILITY

a) insolubility

b) adsorption phenomena

c) gas formation

d) solution pH

1-PHYSICAL INCOMPATIBILITY

- Example 1

  Physical precipitation of Midazolam as a result of unfavorable PH.





1-PHYSICAL INCOMPATIBILITY

- Example 2

❖ Benzyl penicillin and ampicillin lose their potency after 6-8 hrs. if added to dextrose solution due to acidity of the solution (change in pH). Dextrose exhibit catalytic effect on hydrolysis of ampicillin.

2-CHEMICAL INCOMPATIBILITY

a) hydrolysis

b) oxidation reaction

c) reduction reaction

d) photolysis

2-CHEMICAL INCOMPATIBILITY

- Example:

  Chemical precipitation of Midazolam and Ketamine



3-THERAPEUTIC INCOMPATIBILITY

The result of pharmacological effects of several drugs in one patient (drug interactions).

Mechanisms

i. Pharmacokinetic: absorption, distribution, metabolism, excretion (e.g. aminophylline and cimetidine). Cimetidine inhibit metabolism of aminophylline.

ii. Pharmacodynamics: antagonism/ synergism (e.g. Coumadin and Vitamin K)

3-THERAPEUTIC INCOMPATIBILITY

- Example

Heparin and antibiotic as cephalosporins (ceftriaxone),cephalosporin increase the effect of heparin by anticoagulation.

4- DRUG IV CONTAINER INCOMPATIBILITY

Incompatibility that arise from the chemical reaction of the drug and the intravenous container.

a) adsorption: It is the property of a solid/liquid to attract and hold to its surface a gas, liquid, solute, or suspension (e.g. Propofol)

b) Absorption

Some drugs bind to plastic containers and tubing, (e.g.) diazepam and cyclophosphamide

- Some general principles can be applied to the mixing of injection solutions. Mixing is best avoided. If mixing any two or more solutions is necessary, there should be support from published compatibility data.
- A visual check for precipitation, turbidity or color change should be carried out before administering the mixture, but does not guarantee compatibility.

Formula Method

(100 ml for each of the first 10kg) + (50ml for each kg 11-20) + (20 ml for each additional kg) / 24hour

Example:

Calculate the hourly maintenance fluid rate for a child who weighs 25kg

(100mL x 10kg) + (50mL x 10kg) + (20mL x 5kg) / 24hrs

(1000mL) + (500mL) + (100mL) = 1600mL / 24hrs = 66.7ml/hr

Using this formula the hourly fluid maintenance for this child is 67mL/hr

4 / 2 / 1 Method

(4ml/kg for the first 10kg) + (2ml/kg for kg 11-20) + (1ml/kg for every kg above 20) = hourly rate

Example:

Calculate the hourly maintenance fluid rate for a child who weighs 25kg

(4ml x 10kg) + (2ml x 10kg) + (1ml x 5kg) = hourly rate

40ml + 20ml + 5ml = 65ml/hr

Using the 4/2/1 method, this child's hourly maintenance fluid rate is 65mL/hr

The Holliday – Segar 4-2-1 Rule

to estimate Maintenance Hourly Fluid (WATER) Requirements

Weight (kg)	Hourly	Daily
<10 kg	4 mL/kg/hr.	100 mL/kg/day
10 –20 kg	40 mL + 2 mL/kg for every kg >10 kg	1000 mL + 50 mL/kg/day for every kg >10
>20 kg	60 mL + 1 mL/kg for every kg >20 kg	1500 mL + 20 mL/kg/day for every kg > 20

4-2-1 rule EXAMPLES

For a 5 kg infant, Maintenance Hourly Fluid (water) Requirements would be:

4 x 5 = 20ml/hr.

Daily rate: 20 x (24hr) = 480 ml/day

For a 15 kg child, Maintenance Hourly Fluid (water) Requirements would be:

4 x 10 = 40 ml + 2 x 5 = 10 ml
Total: 40 + 10 = 50ml/hr.

Daily rate: 50 x (24hr) = 1200 ml/day

- **“4-2-1 Rule”**
  - 4 ml/kg/hr for the first 10 kg of body weight
  - 2 ml/kg/hr for the second 10 kg body weight
  - 1 ml/kg/hr subsequent kg body weight

Eg : 70 Kg pt

Maintenance fluid : 40+20+50= 110 ml/hr